GRADE 3 · DEBUGGING

What Is a Bug?

Ontario Math C3.2 — read, find, and fix code

Name: ______________________ Date: ______________________

★ **I can find a bug in a loop and fix the repeat number.**

A bug is a mistake in code

A bug is a mistake that makes code do the wrong thing. Debugging is finding and fixing it.

Use 3 steps: say what the code SHOULD do, see what it DOES, then FIX the wrong part. Today the bug is the repeat number.

How to debug

1. SHOULD what it must do → **2. DOES** run & compare → **3. FIX** change the wrong part

Debug in 3 steps: SHOULD → DOES → FIX.

Bit should JUMP 5 times — but this is buggy

```
when green flag clicked
repeat 3
    jump
```

It should be 5 jumps, but repeat 3 makes only 3. Fix the repeat number to 5.

1 Find the bug

Bit should CLAP 6 times. The code says repeat 4. What is wrong, and what should the repeat number be? ______

____

2 Too many this time

Bit should MOVE 4 times, but the code says repeat 7. Is that too many or too few? What should it be?

3 Write the fix

Bit should HOP 8 times. The code says repeat 2. Write the repeat number that fixes it: repeat ______

4 Challenge

Bit should clap 6 times. The loop has TWO claps inside and says repeat 6 — so Bit claps 12! What repeat number gives exactly 6 claps? Explain.

Big idea

To debug: say what the code SHOULD do, compare it to what it DOES, and change the wrong part. A wrong repeat number is the most common loop bug.

TEACHER GUIDE · GRADE 3 · DEBUGGING

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Quick facilitation notes & answer key

What this worksheet teaches

Students learn the debugging method — SHOULD, DOES, FIX — and use it to fix a wrong repeat number in a loop. The method here is SHOULD → DOES → FIX: say what the code SHOULD do, run it to see what it DOES, then FIX the one thing that is wrong. No coding experience needed.

Before you start

No computer needed — the worksheet shows the code and what Bit should do. Optional: a Scratch-style app (for the block loops) or a free Python-turtle playground (for the shapes) to run the code — you only press Run.

How to lead it (about 30 minutes)

1. Show it (you do). Say what Bit SHOULD do (clap 6 times), run the loop to see what it DOES (claps 4), then FIX the repeat number.

2. Do one together. Exercise 1 — change repeat 4 to repeat 6.

3. Let them work. Students fix too-many and too-few repeats (Ex 2–3), and separate total from repeat count when two blocks are inside (Ex 4).

Answer key (these are the verified correct answers)

Exercise 1 — repeat 6 (repeat 4 is too few). Exercise 2 — too many; should be repeat 4.

Exercise 3 — repeat 8. Exercise 4 (challenge) — repeat 3 (2 claps × 3 = 6); repeat 6 would give 12.

If students get stuck — and ways to adjust

Watch for: skipping the SHOULD step. Name the goal first, then compare.

Make it easier: count the goal, count what the loop does, and find the difference.

Make it harder: ask the repeat number when three blocks are inside and Bit should do 9.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

In plain terms: students write and run step-by-step instructions — including a "repeat" instruction — to make something happen.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, find what is wrong, fix it, and explain what the change did.

You'll know it worked when

The child uses SHOULD–DOES–FIX to set the repeat number so the loop matches the goal.